

Formal Verification of Lightning Network Multi-Hop HTLC Routing in Tamarin

A modular, machine-checked model with security analysis and
an empirical study of the prover’s tractability boundaries

Anonymous Author

July 15, 2026

Abstract

We formalise an abstraction of Lightning Network multi-hop HTLC (Hash Time-Locked Contract) routing in the Tamarin symbolic prover under a Dolev–Yao network adversary. The artefact is a *modular package of five theories* totalling **58 machine-checked lemmas**. It covers a two-party channel lifecycle, fixed three-hop signed routing, relational and block-level timing abstractions, and finite-clock witnesses. The principal guarantees are *conditional*: payment atomicity and intermediary safety rely on explicit online-action assumptions, while conservation concerns symbolic routed HTLC amounts rather than the balances of Bitcoin commitment outputs. Within this scope, we (i) replace a global forwarding-replay axiom with a structural linear-resource guard; (ii) machine-check the known *wormhole* fee-theft attack, both as a reachability result and as an equality between the skipped amount and the symbolic path-fee difference; (iii) prove routed-amount equations and expose a griefing DoS; and (iv) demonstrate empirically that the intermediary-claim assumption T3 is load-bearing. A recurring theme is *tractability*: in the tested configuration, signing + natural-number induction + an unbounded block clock exhausts the prover under our resource bounds. We also map two further empirical boundaries—signed-message N -hop routing and reusable channels—where automated search stops scaling. All reported prover verdicts and bounded timeouts were executed under the commands and limits stated in the artefact.

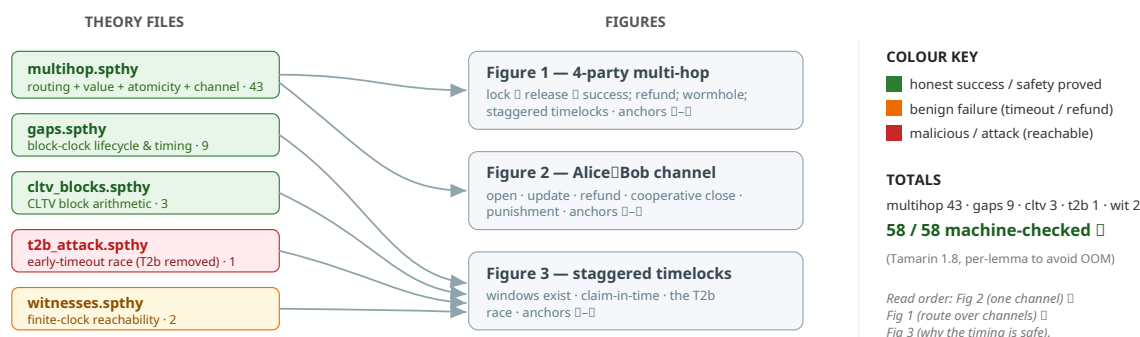
Contents

1	Introduction and scope	2
2	Architecture: a modular five-theory package	3
3	Related work	4
4	Methodology	4
5	Results: the core model	5
5.1	Multi-hop routing (<code>multihop.spthy</code>)	5
5.2	Timing safety (<code>cltv_blocks</code> , <code>gaps</code> , <code>t2b_attack</code>)	5
5.3	Statistics	8

6	Security analysis	8
6.1	Forwarding-replay hardening (axiom to structure)	8
6.2	Wormhole attack (reachable and quantified)	8
6.3	Economic soundness and griefing	9
6.4	Assumption minimality (T3 is load-bearing)	9
7	Generalisation to N hops	9
8	What failed, and the tractability boundaries	10
9	Reproduction	10
10	Conclusion	11

Verification map — 5 core theories · 3 protocol views + master map · 58 lemmas

where each file's lemmas live, and how the figures fit together



The package also includes a separate idealised N-hop abstraction (multihop_nhop_fees.spthy, 31/31) and the reusable-channel boundary spike (experiments/), documented in REPORT2.md.

Read this map first, then the routing, channel-lifecycle, and timing views.

Figure 1: Verification map: five core theory files, three protocol views (plus this master map), and 58 lemmas. Green = honest success / safety proved; amber = benign failure (timeout / refund); red = malicious / attack shown reachable.

1 Introduction and scope

Lightning is a Layer-2 payment-channel network: parties transact off-chain and use the blockchain only to settle disputes. A multi-hop payment routes value along a path Alice → Bob → Carol → Dave by chaining HTLCs that share one payment hash $y = H(x)$; the receiver reveals the preimage x , which propagates back along the path, redeeming each HTLC. Our reference model is the standard textbook presentation [9]: a four-party path, per-hop fees, staggered CLTV timelocks $3t/2t/t$, and a two-round “lock then release” commit. The Lightning specification itself is maintained as the BOLT documents [4].

Threat model. The adversary controls the symbolic network: it may intercept, replay, synthesize, and reroute any derivable message, and may reveal long-term signing keys through explicit compromise rules. Cryptographic primitives are perfect modulo Tamarin’s equational theories. The adversary does not break hashes or signatures, control Bitcoin consensus, or violate the trace restrictions stated below.

Scope and limitations. We model integrity and trace safety, not payment privacy or unlinkability (which require onion routing and observational-equivalence proofs). The five theories are separate labelled transition systems: their results support one another conceptually, but no formal refinement theorem transfers a lemma from one theory to another. In the shipped fixed-hop model, the linear `Free(ptr)` guard is consumed and not returned, so a channel hosts one HTLC over its modelled lifetime. HTLC amounts are related by symbolic natural-number equations but are not debited from the `StateA/StateB` channel balances; “value conservation” below therefore means routed-amount conservation, not ledger solvency. Timing is relational in the main theory, while separate theories show that numerical CLTV claim windows exist. They do not prove that an online party acts; T2b/T3 state that liveness boundary explicitly.

Contributions. Within those limits, the artefact contributes: (1) a signed fixed-hop Dolev–Yao model with an explicit assumption matrix; (2) structural replay hardening using a linear resource rather than a global single-spend axiom; (3) machine-checked reachability of known wormhole and grieving behaviours; (4) an idealised linear-fact generalisation of routing safety to arbitrary path length; and (5) reproducible positive and negative results characterising where Tamarin’s automated search ceases to scale.

Artefact. Sources, figures, and the runner are available at <https://github.com/AAlilbitme/12345>. The core Stage 1 model corresponds to commit d78740a; the reusable-channel experiment to 2700974. Exact commands appear in Section 9.

2 Architecture: a modular five-theory package

A single monolithic theory is intractable: combining Tamarin’s signing equational theory, natural-number induction, and an unbounded block clock exhausted memory in the tested configuration. Pairwise experimental variants remained tractable under the recorded bounds. This is empirical tool evidence, not a general complexity claim. The model is therefore split so that no single core theory carries all three (Table 1).

File	Aspect	Lemmas	Builtins
<code>multihop.spthy</code>	lifecycle + routing + value/fees + atomicity	43	hashing, signing, nat
<code>gaps.spthy</code>	block-clock lifecycle & CLTV timing safety	9	natural-numbers
<code>cltv_blocks.spthy</code>	pure CLTV block arithmetic	3	natural-numbers
<code>t2b_attack.spthy</code>	early-timeout race (liveness assumption removed)	1	—
<code>witnesses.spthy</code>	finite-clock reachability witnesses	2	natural-numbers
Total		58	

Table 1: The five theories. `multihop.spthy` combines signing + nat once the unbounded clock is dropped and timing is kept relational—which is why value/fee conservation could be merged into it.

3 Related work

Tamarin performs symbolic trace reasoning over multiset-rewriting rules and equational theories [1, 2]. Boyd, Gjøsteen, and Wu model a blockchain and an HTLC-based cross-chain protocol in Tamarin [5]; their ledger and global-time treatment is richer than the relational timing abstraction in our main theory. Hüttel and Staroveški use ProVerif to study secrecy and authenticity of four Lightning subprotocols [6]. Grundmann and Hartenstein model check Lightning in TLA+ using explicit refinement arguments to separate channel and multi-hop reasoning [7]; in contrast, our modular split is a tractability device without a machine-checked refinement between theories. Malavolta et al. identify the wormhole attack and propose anonymous multi-hop locks [8]; our contribution is not discovery of that attack, but a Tamarin witness and symbolic fee-difference characterization in this model. The original Lightning proposal and the BOLT specifications provide the protocol context [3, 4].

4 Methodology

All results were produced under a strict *test-first* discipline: a property is claimed only after Tamarin reports it, and any “plausible” existence result is re-checked with a tightened variant before being trusted (Section 8 shows this catching a false positive). Restrictions (global trace filters) are treated as *assumptions*, not proofs; a suspiciously short proof is a warning sign. Arithmetic theories motivate the timing orderings, while online action remains an explicit liveness assumption. Likewise, a timeout is a measured negative tool outcome under a stated command and wall-clock bound, not a theorem that no proof or witness exists.

Per-lemma invocation (rather than one bulk `-prove`) is used on the large theory to avoid OOM:

```
LANG=C.utf8 LC_ALL=C.utf8 tamarin-prover multihop.spthy \
--heuristic=c --stop-on-trace=seqdfs --derivcheck-timeout=0 --prove=<Lemma>
```

The flag `-stop-on-trace=seqdfs` rescues the heavy existence witnesses but *hangs* the natural-number arithmetic theories, so the runner applies it per file and lemma. The whole suite is driven by `run.py`; the `Makefile` is a thin wrapper.

Claim family	Theory / mechanism	Assumptions or limits
Fixed-hop authentication and causality	<code>multihop.spthy</code> ; signatures and linear pending facts	Three fixed hops; perfect cryptography; one HTLC per channel lifetime
Atomicity and intermediary safety	<code>multihop.spthy</code> ; T1/T2/T2b/T3 restrictions	Conditional on timely online action; short proofs mostly discharge restrictions
Routed-amount equations	<code>multihop.spthy</code> ; natural-number action equations	Not tied to commitment balances or capacity
Post-close exclusion and CLTV windows	<code>gaps.spthy</code> , <code>cltv_blocks.spthy</code>	Separate abstract transition systems; no refinement into the main model
Arbitrary-length routing safety	<code>multihop_nhop_fees.spthy</code> ; linear Route/Fulfill facts	Idealised internal channel; no signed forwarding-message adversary

Table 2: Assumption and abstraction matrix. A Tamarin verdict is interpreted relative to the theory and assumptions in its row.

5 Results: the core model

5.1 Multi-hop routing (multihop.spthy)

Figure 2 maps the routing, value, and attack lemmas onto the four-party flow; Figure 3 maps the channel-lifecycle lemmas onto a single Alice↔Bob channel. All 43 lemmas verify.

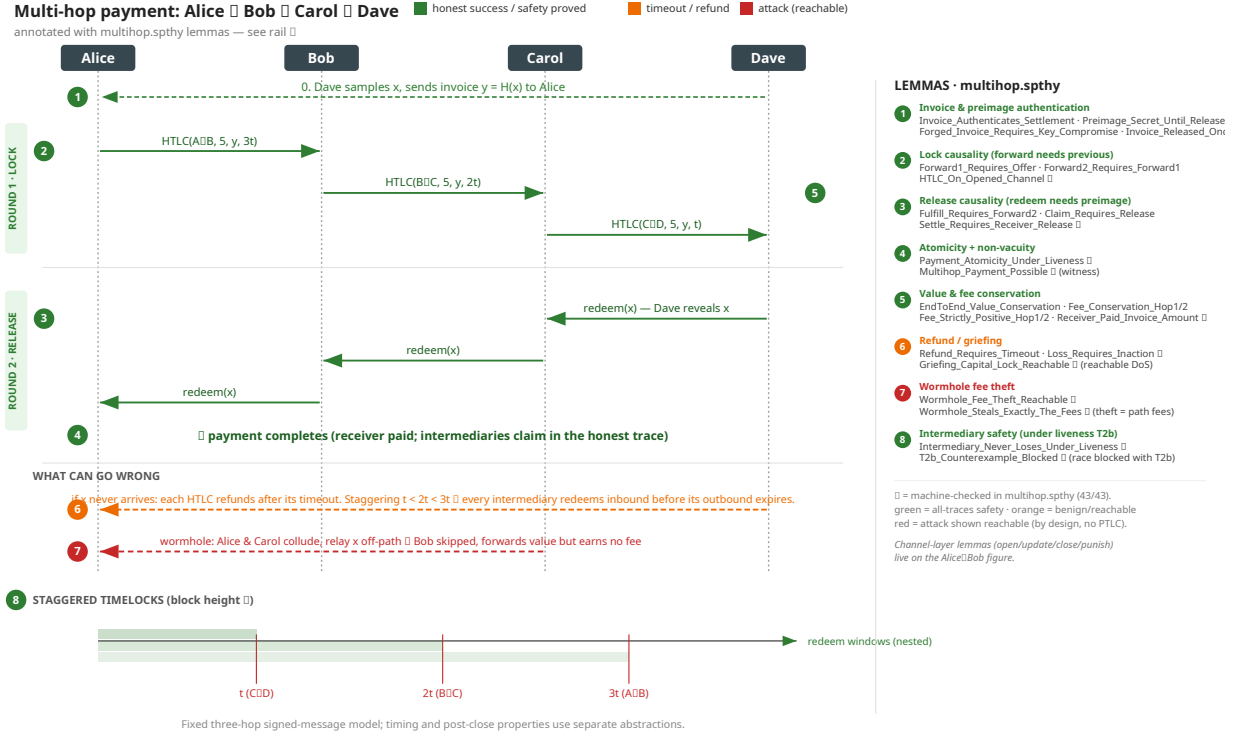
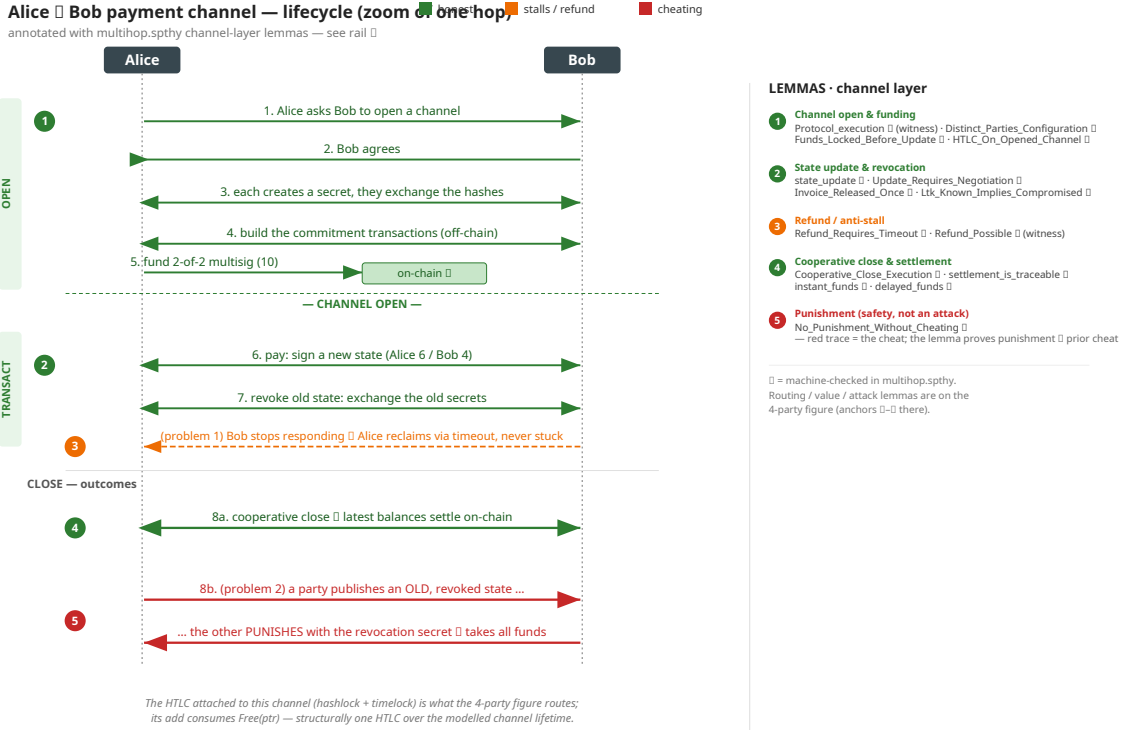


Figure 2: Four-party multi-hop payment. Round 1 locks HTLCs forward with staggered timelocks $3t/2t/t$; Round 2 releases the preimage backward. Numbered anchors 1–8 key to the `multihop.spthy` lemma rail. The red line is the wormhole; the amber line the timeout/refund unwind.

5.2 Timing safety (cltv_blocks, gaps, t2b_attack)

Figure 4 shows the staggered-timelock story. `cltv_blocks` proves arithmetic consequences of explicit deadline restrictions—in particular that the assumed stagger gives a non-empty claim window. `gaps` proves lifecycle properties in a *different*, linear-channel/block-clock abstraction (including no HTLC addition after close) and provides redeem/refund witnesses. These lemmas do not automatically hold in `multihop.spthy`. Finally, `t2b_attack` is a minimal over-approximating theory that exhibits the early-timeout ordering possible without T2b; it is not literally the main theory with one restriction deleted.



Channel-layer lemmas in the fixed-hop abstraction; routing and timing are shown separately.

Figure 3: The Alice↔Bob channel lifecycle (a zoom of one hop): open, state update, refund, cooperative close, and punishment. Note anchor 5: the red trace is the *cheat*, but the lemma `No_Punishment_Without_Cheating` is a safety guarantee (punishment $\Rightarrow$ prior cheat).

Group	Class	Representative lemmas
Invoice / preimage auth	■	Invoice_Authenticates_Settlement, Preimage_Secret_Until_Released, Forged_ Invoice_Requires_Key_Compromise
Lock causality	■	Forward1_Requires_Offer, Forward2_ Requires_Forward1, HTLC_On_Opened_ Channel
Release causality	■	Fulfill_Requires_Forward2, Claim_ Requires_Release, Settle_Requires_ Receiver_Release
Atomicity / non-vacuity	■	Payment_Atomicity_Under_Liveness, Multihop_Payment_Possible
Value & fees	■	EndToEnd_Value_Conservation, Fee_ Conservation_Hop1/2, Fee_Strictly_ Positive_Hop1/2, Receiver_Paid_Invoice_ Amount
Refund / griefing	■	Refund_Requires_Timeout, Loss_Requires_ Inaction, Griefing_Capital_Lock_ Reachable
Wormhole	■	Wormhole_Fee_Theft_Reachable, Wormhole_ Steals_Exactly_The_Fees
Intermediary safety	■	Intermediary_Never_Loses_Under_ Liveness, T2b_Counterexample_Blocked
Channel lifecycle	■	state_update, Protocol_execution, Cooperative_Close_Execution, No_ Punishment_Without_Cheating, Funds_ Locked_Before_Update

Table 3: Lemma groups in `multihop.spthy` (43 total). Class: ■ safety proved, ■ benign/reachable, ■ attack reachable.

Lemma (representative)	Steps	Time	Kind
Multihop_Payment_Possible	70	~77 s	exists-trace (heaviest)
Forward2_Requires_Forward1	274	~47 s	all-traces
Forward1_Requires_Offer	266	~46 s	all-traces
Wormhole_Steals_Exactly_The_Fees	81	~45 s	exists-trace
Wormhole_Fee_Theft_Reachable	51	~40 s	exists-trace
Intermediary_Never_Loses_Under_Liveness	2	<1 s	all-traces (under T2b)
EndToEnd_Value_Conservation	2	<1 s	all-traces

Table 4: Representative per-lemma statistics for `multihop.spthy` (43 lemmas verify per-lemma sequentially in ≈ 16 min total on the test machine; average ≈ 26 s; many trivial lemmas finish in <1 s).

Staggered CLTV timelocks — timing safety (block height)

covers `cltv_blocks.spthy` · `gaps.spthy` · `t2b_attack.spthy` — see rail

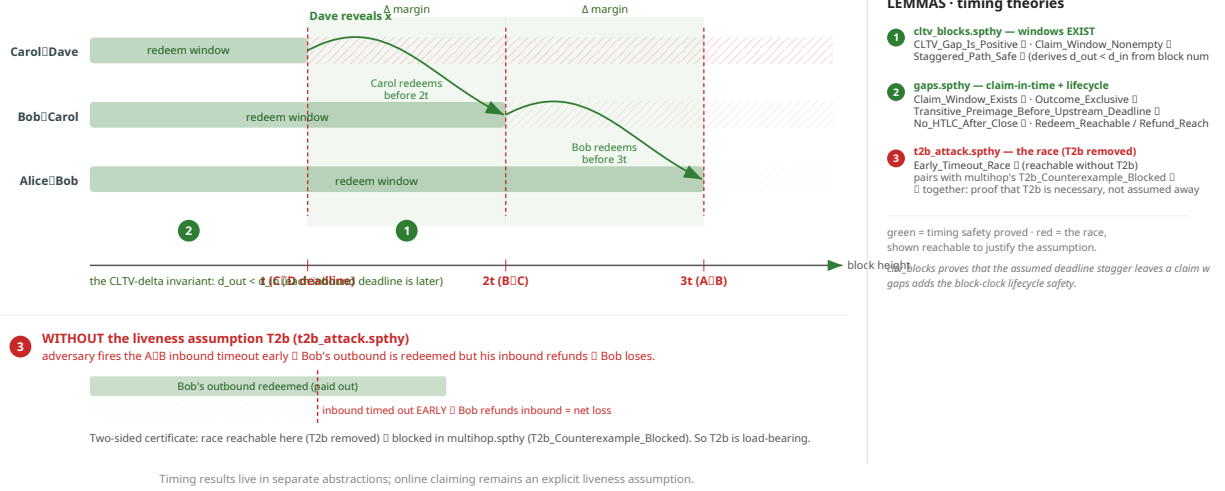


Figure 4: Staggered CLTV timelocks in the timing abstractions. The arithmetic model shows a Δ margin when every inbound deadline is later than its outbound one. Bottom (red): a minimal theory without T2b admits an ordering in which the inbound timeout precedes the outbound redemption (`Early_Timeout_Race`). Together with the blocked formula in the main model, this is evidence that T2b is load-bearing, not a cross-theory refinement proof.

5.3 Statistics

6 Security analysis

6.1 Forwarding-replay hardening (axiom to structure)

The forwarding rules read the outgoing amount/fee from the network and are gated only by a public HTLC-offer message. A Dolev–Yao replay of that offer could re-fire an honest forward, locking a *second* HTLC on the same channel output with an adversary-chosen amount. This was found empirically via an audit probe (double-forward: reachable) and initially blocked with a global axiom `OneHTLCPerPtr`. We then replaced the axiom with a *structural* guard: each channel open emits one linear `Free(ptr)` token, and every HTLC-add consumes it. “At most one HTLC per output” is now a consequence of linear-fact consumption—one fewer assumption for the same guarantee. After the change the double-forward probe is falsified structurally (100 steps, no axiom) and all 43 lemmas still verify. We also confirmed that a symmetric “double-redeem” is *not* a vulnerability: the redeem/settle rules each consume a linear pending fact, so replay cannot double-spend an output.

6.2 Wormhole attack (reachable and quantified)

The wormhole attack (Malavolta et al., NDSS’19): two colluding path members relay the preimage out-of-band, skipping an honest intermediary who forwards value but earns no fee [8]. In the symbolic model the public Dolev–Yao network over-approximates such a side channel; collusion and final account ownership are not represented explicitly. We machine-check it twice: `Wormhole_Fee_Theft_Reachable` (51 steps) exhibits a trace where the payment settles yet the middle hop is

never redeemed and no intermediary earns a fee, with *no* key compromise; and **Wormhole_Steals_Exactly_The_Fees** (81 steps) shows that the symbolic sender/receiver amount difference equals $%f_1\% + %f_2$ while neither intermediary emits **FeeEarned**. Because channel balances and transfers to colluders are not modeled, this is a skipped-fee characterization rather than a ledger-level proof of who receives stolen coins. Anonymous multi-hop locks address the known attack class [8].

6.3 Economic soundness and grieving

The routed-amount lemmas are direct consequences of equations emitted by the forward rules. **EndToEnd_Value_Conservation** composes the two per-hop equations, but does not constrain the separate channel-state balances. **Fee_Strictly_Positive_Hop1/2** follows because this encoding uses Tamarin’s positive-natural sort (there is no $%0$); it therefore records a modeling choice and should not be read as a claim that Lightning forbids zero-fee channels. As a reachability result, **Griefing_Capital_Lock_Reachable** shows a trace labeled as capital lock/refund with no completed payment, abstracting the standard PCN grieving DoS.

6.4 Assumption minimality (T3 is load-bearing)

A natural question is whether the intermediary-claim restriction T3 is redundant given T1 (redeem-before-timeout) and T2b (honest parties act before the inbound timeout). We tested it by removing T3 and re-running the suite: the safety lemma **Intermediary_Never_Loses_Under_Liveness** is then *falsified*, Tamarin finding a concrete 59-step counterexample. T2b orders the two timeouts but does not force the node to *sweep* the incoming HTLC once its outgoing one is redeemed; that action is exactly what T3 encodes. This establishes that T3 is load-bearing relative to the remaining assumptions. It does *not* establish independent minimality of every timing restriction; doing so would require one removal experiment and reproducible counterexample per restriction.

7 Generalisation to N hops

The core model fixes the path at three hops. Replacing the two hardcoded forward rules with one *generic* rule was attempted in several encodings, all run:

Encoding	Outcome	
Signed-message generic forward	self-referential signed term; origin analysis re-curses, even a one-hop witness OOMs	■
+ fresh per-hop id	fixes replay but <i>not</i> termination—still a signed term producible only by a forward	■
Linear-fact routing (idealised channel)	safety invariants verify by induction over the chain, any N	■

Table 5: N -hop encodings. The bounding factor is cryptographic origin-analysis recursion on self-referential signed messages, not path length.

The linear-fact route yields **multihop_nhop.spthy** (28/29 lemmas; the one open lemma, **Settle_Requires_Receiver_Release**, keys on the terminal settle and does not terminate) and, with amounts/fees added, **multihop_nhop_fees.spthy** (31/31, verified in 2 min 02 s on the test machine). The latter signs the invoice but represents forwarding and fulfilment by internal linear facts rather than network messages. It is therefore an idealised routing-state abstraction, not a

cryptographic generalisation of the fixed-hop model. The two layers have no formal refinement proof.

8 What failed, and the tractability boundaries

We record negative results explicitly—they are the paper’s methodological contribution as much as the positive ones.

- **The three-ingredient OOM.** Signing + natural-number induction + an unbounded block clock together exhausted the tested configuration. Selected pairwise variants remained tractable; the artefact should be read as empirical evidence under the stated resource limits, not a general complexity theorem. This motivates the modular split (Section 2).
- **Signed-message N -hop routing** (Table 5): origin analysis on the self-referential `'htlc'` term does not terminate, even for reachability, even with a fresh id.
- **A rejected restriction.** An initial fix `OneRedeemPerPtr` (“at most one redeem per output”) was tested and *rejected*: it falsified the honest witness `Multihop_Payment_Possible`, because the two `Redeemed` events on the sender’s output in an honest run are the two endpoints’ views of one hop, not two spends. The correct guard was `OneHTLCPerPtr` (later made structural).
- **A caught false positive.** An exists-trace probe that shared only the payment *hash* between an offer and a payment appeared to show “value creation” (receiver paid more than the sender locked, no compromise). Tightening it to a single *linked* payment chain *falsified* it, while a control confirmed the chain is otherwise reachable—so the apparent attack was invoice-hash reuse across two unrelated offers, not a soundness bug. This is the adversarial-verification discipline in action.
- **Reusable channels (boundary).** We staged a refactor toward faithful reusable channels: Stage 1 (structural single-HTLC-per-output via the `Free(ptr)` slot) ships and re-verifies 43/43. Stage 2 (return `Free(ptr)` on resolution, so a channel carries sequential HTLCs) was spiked and is a clean tractability boundary: without `-auto-sources` every lemma returns “analysis incomplete” (the reuse loop leaves unbounded partial deconstructions); *with* `-auto-sources` the backward-reasoning safety lemmas verify (~ 45 s each), but every exists-trace witness and the interval lemma `No_Concurrent_HTLC_Per_Output` fail to converge. The original interval formula was unguarded; after splitting redeem/refund into two guarded existential disjuncts, the theory passes `-quit-on-warning`, but the corrected lemma still produced no result after approximately 786 s. A tightly constrained sender-only reuse witness timed out at 900 s under both default and SEQDFS search. Reusable channels are expressible and their core safety survives, but the loop reintroduces the witness/liveness non-termination the modular split avoids; Stage 1 is therefore the shipped model.

9 Reproduction

```
export LANG=C.utf8 LC_ALL=C.utf8
python3 run.py # full core suite (5 theories)
python3 run.py multihop.spthy # one theory
make nhop # N-hop fee abstraction (31/31)
```

The recorded core results use Tamarin 1.8 and Maude 3.1. The N-hop-fee suite was additionally re-run with Tamarin 1.12.0 and Maude 3.5 (31/31); step counts may differ across versions. The Makefile delegates to `run.py`, which selects per-file strategies and returns nonzero if a lemma is falsified, incomplete, missing, or timed out. The Stage 2 experiment requires `-auto-sources`; its timeout observations use the explicit limits reported above.

10 Conclusion

We delivered a modular, machine-checked *abstraction* of Lightning multi-hop HTLC routing (58 lemmas over five separate theories). The fixed-hop model replaces a replay-prevention axiom with a linear structural guard, reproduces known wormhole and griefing behaviours, and verifies authentication, causality, and routed-amount equations under explicit assumptions. Separate theories establish arithmetic claim windows, post-close exclusion in a linear channel abstraction, and finite-clock witnesses. These results do not constitute a unified balance-correctness proof of the BOLT protocol: channel-balance integration, formal refinement between theories, privacy, concurrent HTLCs, and unbounded reusable channels remain open. The durable contribution is methodological and empirical—a reproducible account of which abstractions keep Tamarin tractable, which assumptions carry the conditional guarantees, and where signed generic routing and reusable-state witness search stop scaling.

References

- [1] B. Schmidt, S. Meier, C. Cremers, and D. Basin. Automated analysis of Diffie–Hellman protocols and advanced security properties. In *25th IEEE Computer Security Foundations Symposium (CSF)*, pages 78–94, 2012.
- [2] S. Meier, B. Schmidt, C. Cremers, and D. Basin. The Tamarin prover for the symbolic analysis of security protocols. In *25th International Conference on Computer Aided Verification (CAV)*, 2013.
- [3] J. Poon and T. Dryja. The Bitcoin Lightning Network: Scalable off-chain instant payments. Technical report, 2016.
- [4] Lightning Network specification contributors. Basis of Lightning Technology (BOLTs). <https://github.com/lightning/bolts>, accessed July 2026.
- [5] C. Boyd, K. Gjølsteen, and S. Wu. A blockchain model in Tamarin and formal analysis of Hash Time Lock Contract. In *2nd Workshop on Formal Methods for Blockchains (FMBC)*, OASICS 84, 2020. DOI: 10.4230/OASICS.FMBC.2020.5.
- [6] H. Hüttel and V. Staroveški. Secrecy and authenticity properties of the Lightning Network protocol. In *6th International Conference on Information Systems Security and Privacy (ICISSP)*, pages 119–130, 2020.
- [7] M. Grundmann and H. Hartenstein. Security of the Lightning Network: Model checking a stepwise refinement with TLA+. In *20th International Conference on Integrated Formal Methods (iFM)*, 2025. DOI: 10.1007/978-3-032-10794-7_16.
- [8] G. Malavolta, P. Moreno-Sanchez, A. Schneidewind, A. Kate, and M. Maffei. Anonymous multi-hop locks for blockchain scalability and interoperability. In *Network and Distributed System Security Symposium (NDSS)*, 2019.
- [9] M. Maffei. Cryptocurrencies, Lecture 11: Scalability. TU Wien lecture slides, December 2024.